BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022 • Paper Code: 8463

Retyped faithfully from the original board paper — same questions, no changes.

Biology - Paper: II - Objective (ii)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8463

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. How many codons specify the amino acid leucine?

(A) 2
(B) 4
(C) 6
(D) 3

Q2. During cytokinesis in animal cells, actin and myosin form contractile ring which is then followed by the formation of:

(A) Equatorial plate
(B) Cleavage furrow
(C) Phragmoplast
(D) Cell plate

Q3. During pachytene:

(A) Pairing of homologous chromosomes start
(B) Pairing of homologous chromosomes is completed
(C) Paired chromosomes start to separate
(D) Separation of chromosomes is completed

Q4. What type of ratio 1:2:1 is?

(A) Genotypic ratio of monohybrid cross
(B) Phenotypic ratio of monohybrid cross
(C) Genotypic ratio of dihybrid cross
(D) Phenotypic ratio of dihybrid cross

Q5. Reverse transcription yields:

(A) mRNA
(B) cDNA
(C) rDNA
(D) tRNA

Q6. The use of PROBE is:

(A) To study palindromic sequence
(B) To study bacterial plasmids
(C) To study phage DNA
(D) To search genomic library

Q7. The change in allele frequency which occurs by chance is:

(A) Genetic drift
(B) Emigration
(C) Immigration
(D) Mutation

Q8. The actual location of an organism is called its:

(A) Niche
(B) Habitat
(C) Behavior
(D) Environment

Q9. Succulent plants are found in:

(A) Alpine forests
(B) Grassland
(C) Desert
(D) Deciduous forest

Q10. Which one is an environmental buffer?

(A) Ocean
(B) River
(C) Desert
(D) Forest

Q11. Urine leaves the kidney through:

(A) Urethra
(B) Ureter
(C) Renal pelvis
(D) Ureteral orifice

Q12. Which of these move first leading to the exosmosis of water during rapid movements?

(A) Cl^{-1} ions
(B) NH_4^{+1} ions
(C) K^{+1} ions
(D) Ca^{+2} ions

Q13. To which region of vertebral column, tetrapod's pelvic girdle is attached?

(A) Lumber region
(B) Sacral region
(C) Pelvic region
(D) Cervical region

Q14. Which of these commercially produced hormone promotes malting?

(A) GA_3
(B) GA
(C) 2,4 D
(D) N.A.A.

Q15. Which of these differentiates into mature sperms in human male?

(A) Spermatogonia
(B) Primary spermatocytes
(C) Secondary spermatocytes
(D) Spermatids

Q16. Inhibitory effect of lateral shoots is caused by:

(A) Abscisic acid
(B) Gibberellins
(C) Ethene
(D) Auxins

Q17. Notochord is seen in the chick embryo of:

(A) 24 hrs.
(B) 22 hrs.
(C) 20 hrs.
(D) 18 hrs.

Biology - Paper: II - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any 3 questions from Section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Discuss adaptations of animals in terrestrial ecosystem.

ii. What are excretophore?

iii. Draw the sketch of Urea Cycle.

iv. What is spondylosis?

v. Explain all or none response of muscle fiber.

vi. Discuss locomotion in Mammals?

vii. What are Gonadotrophins? Give their roles in male and female.

viii. Define afterbirth.

ix. Differentiate between alpine and boreal forests.

x. Discuss the soil condition of the grassland ecosystem.

xi. Give different ways of energy conservation.

xii. What is acid rain?

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Differentiate between chemoreceptors and thermoreceptors.

ii. Which receptors respond to the mechanical conditions of the internal organs? Give examples.

iii. What happens when an impulse reaches a synaptic knob?

iv. What is a gene pool?

v. Write about over dominance.

vi. Narrate epistasis.

vii. What are restriction enzymes, who isolated them?

viii. How transgenic bacteria promote health of plants? Give example.

ix. Define examine method of Gene Therapy.

x. Differentiate between population and community.

xi. Write a note on blood components.

xii. How micronutrients differ from macronutrients?

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. What is parthenocarpy? How it can be used commercially?

ii. How fraternal twins are produced?

iii. What is a nucleosome? Why histones are positively charged?

iv. Define transformation. Who discovered this phenomenon?

v. Compare Okazaki fragments of prokaryotes and eukaryotes.

vi. Draw ultrastructural features of cell death by apoptosis.

vii. Give chromosomal make up and symptoms of Turner's Syndrome.

viii. State theory of special creation.

ix. What is meant by endosymbiont hypothesis? Who proposed this hypothesis?

> Note: SECTION-C (EACH QUESTION CARRIES EIGHT (8) MARKS)

Q5.. (Marks 4) (a) Explain excretion in Cockroach with diagram. (b) Discuss important steps of Nitrogen Cycle.

Q6.. (Marks 4) (a) How is human skeleton deformed by trauma? Justify your answer using special reference of disc slip. (b) What are mutations? How are they classified?

Q7.. (Marks 4) (a) How is growth affected by STH, thyroxine and adrenal hormones of human endocrine system? (b) Explain the reasons of Ozone layer depletion.

Q8.. (Marks 4) (a) Describe the process of birth in human female. (b) What is diabetes mellitus? Discuss diabetes type I disease.

Q9.. (Marks 4) (a) What is differentiation? Explain its phenomenon in the formation of different structure of the body. (b) How Prokaryotes evolved into Eukaryotes? Discuss the different hypothesis in this type of evolution.